

Structural Organisation in Animals

1. From Cell to Organ System

Unicellular organisms: A single cell performs all functions — digestion, respiration, excretion, reproduction, etc.

Example: Amoeba.

Multicellular organisms: Functions are divided among different groups of cells.

These groups form tissues, which form organs, which combine to make organ systems.

This is called division of labour.

Hierarchy of Organisation

Level — Organisation

Cell Basic structural and functional unit of life

Tissue Group of similar cells + intercellular substance performing a common function

Organ Structure composed of two or more types of tissues

Organ system Group of organs working together for a common function

Example:

Organ: Heart (contains epithelial, connective, muscular & neural tissues)

Organ system: Circulatory system.

2. Four Basic Types of Animal Tissues

Epithelial Tissue — covers body surface & lines cavities.

Connective Tissue — supports and binds other tissues.

Muscular Tissue — allows movement.

Neural Tissue — controls & coordinates body functions.

1. Epithelial Tissue

Epithelial tissue covers the body surfaces, lines internal cavities, and forms glands. It provides protection, secretion, absorption, and sensory reception.

There are different types of epithelial tissues:

Simple squamous epithelium consists of a single layer of flat, thin cells. It is found in alveoli of lungs and lining of blood vessels, where it helps in diffusion and filtration.

Simple columnar epithelium has a single layer of tall, column-like cells. It is present in the lining of the stomach and intestine, functioning in secretion and absorption.

Stratified squamous epithelium consists of many layers of flat cells. It forms the outer layer of skin and lines the oral cavity and esophagus, serving a protective function.

Ciliated epithelium is composed of columnar cells bearing hair-like cilia on their surface. This type is found in the trachea and bronchi, where it moves mucus and trapped particles.

Glandular epithelium consists of cuboidal or columnar cells that form secretory units of glands, such as the salivary glands and pancreas, and is specialised for secretion.

2. Connective Tissue

Connective tissue binds, supports, and protects other tissues and organs. It also transports substances and stores energy.

Examples include:

Loose connective tissue, – where cells are scattered in a matrix of collagen and elastic fibres, found beneath the skin and between muscles for support and binding.

Adipose tissue, – composed of cells filled with fat droplets, located beneath the skin and around organs like the heart and kidneys for insulation and energy storage.

Dense connective tissue, – with densely packed collagen fibres, found in tendons and ligaments, providing strong attachment between muscles and bones.

Cartilage, – with a firm but flexible matrix, present in structures like the nose, ear, and joints, offering support and flexibility.

Bone, – having a hard matrix containing collagen fibres, forms the skeleton and provides structural support and protection.

Blood, – a liquid connective tissue with plasma as matrix, circulates through the body to transport gases, nutrients, and hormones.

Muscular Tissue:

Muscular tissue is specialised for movement. It has the ability to contract and relax, enabling movement of body parts and internal organs.

Types:

Skeletal muscle – long, cylindrical, striated, multinucleated fibres; attached to bones; responsible for voluntary movement.

Cardiac muscle – branched, striated, uninucleated fibres with intercalated discs; found in heart; responsible for pumping blood involuntarily.

Smooth muscle – spindle-shaped, non-striated cells; found in walls of stomach, intestine, bladder; responsible for involuntary movement of internal organs.

Function: Movement of body and internal organs.

4. Neural Tissue

Neural tissue is specialised for communication and coordination. It transmits electrical impulses and controls body functions.

A neuron, the structural unit of neural tissue, consists of:

Cell body (soma) containing the nucleus and cytoplasm.

Dendrites, branched extensions that receive impulses from other neurons.

Axon, a long fibre that transmits impulses away from the cell body.

Myelin sheath, an insulating covering on the axon that speeds

Neural tissue is found in the brain, spinal cord, and nerves. Its main functions include receiving stimuli, conducting impulses, and coordinating responses.

3. Organ Systems

Organs work together in a specific pattern to perform complex functions.

Example:

Digestive system → mouth, esophagus, stomach, intestines, etc.

Respiratory system → lungs, trachea, bronchi.

Importance: Division of labour ensures efficiency and coordination.

Morphology and Anatomy of Frog.

Morphology:

Study of form or externally visible features of an organism.

For animals → shape, size, colour, external body structure.

Example: In frog → study of body shape, limbs, skin, eyes, mouth.

Anatomy:

Study of internal structure of animals — arrangement of organs and systems.

Example: In frog → study of internal organs such as heart, lungs, liver, digestive tract.

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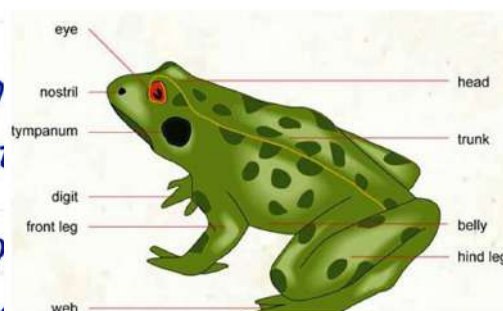
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Internal Features of Frog (Anatomy):

Digestive system: Mouth → pharynx → oesophagus → stomach → small intestine → large intestine → cloaca.

Circulatory system: Three-chambered heart (two atria, one ventricle).

Respiratory system: Lungs + skin (cutaneous respiration).

Excretory system: Kidneys, ureters, urinary bladder.

Nervous system: Brain + spinal cord.

Reproductive system: Testes or ovaries depending on sex.

Digestive System

Consists of alimentary canal and digestive glands.

Short alimentary canal as frogs are carnivores (short intestine).

Path of food: Mouth → Buccal cavity → Pharynx → Oesophagus → Stomach → Intestine → Rectum → Cloaca.

Special features:

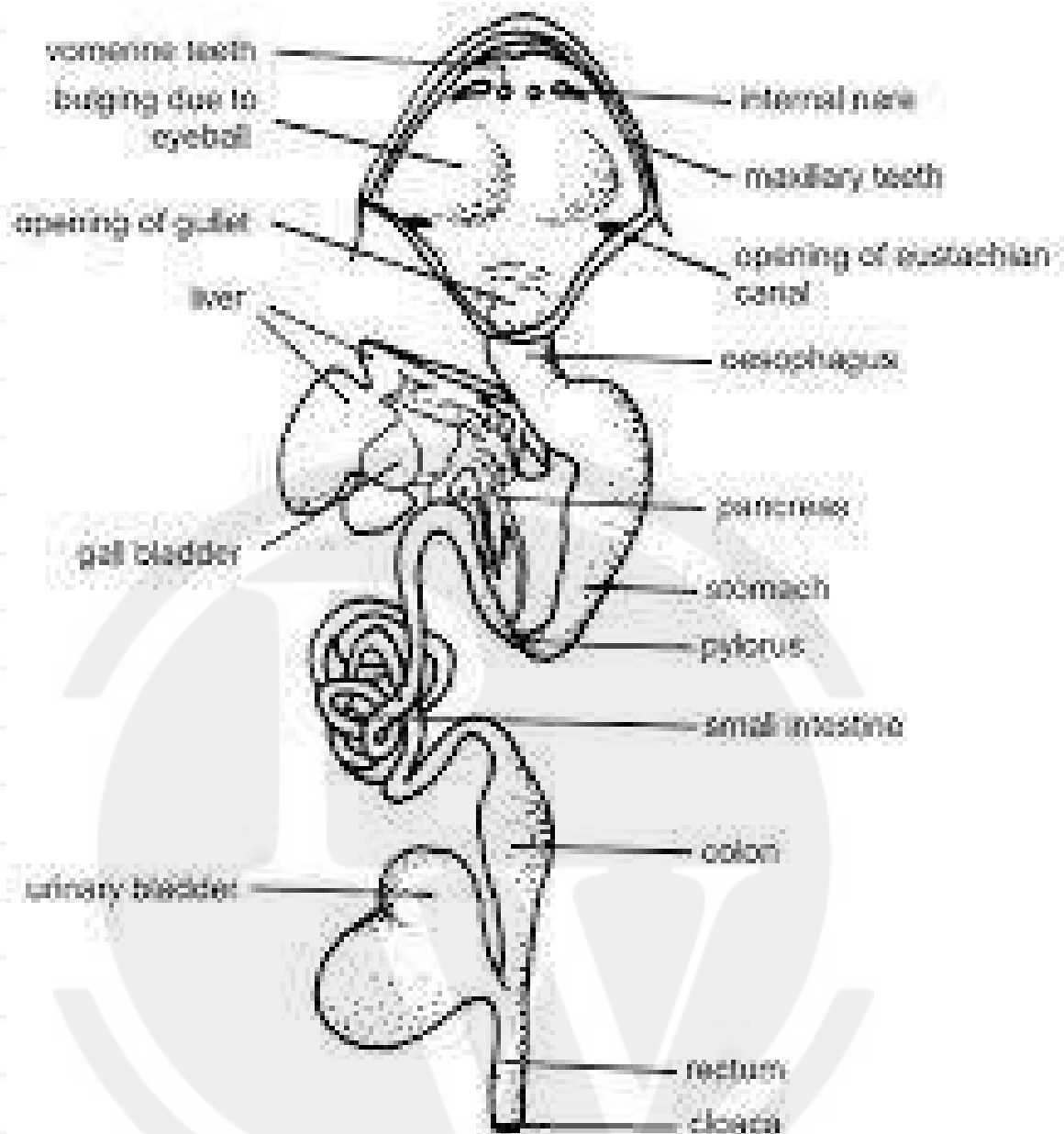
Bilobed tongue to capture food.

Stomach secretes HCl and gastric juices for digestion.

Chyme moves to the duodenum (first part of small intestine).

Duodenum receives bile from gall bladder and pancreatic juice via common bile duct.

Bile emulsifies fats; pancreatic juice digests carbohydrates and proteins.



Final digestion occurs in the intestine.

Absorption via villi and microvilli in intestine.

Undigested waste moves to rectum → cloaca for excretion.

Digestive glands:

Liver: secretes bile (stored in gall bladder).

Pancreas: produces digestive enzymes.

2. Respiratory System

Frogs respire by cutaneous respiration in water and pulmonary respiration on land.

In water: Skin acts as respiratory organ; gaseous exchange by diffusion.

On land: Buccal cavity, skin, and lungs are respiratory organs.

Lungs: Paired, elongated sac-like structures in the thorax.

Air pathway: Nostril → Buccal cavity → Lungs.

During hibernation/aestivation: Respiration occurs through skin.

3. Circulatory System

Type: Closed and well-developed; also has lymphatic system.

Components: Heart, blood vessels, blood.

Heart:

Muscular structure in the upper body cavity.

Three chambers: two atria, one ventricle.

Covered by pericardium

Covered by pericardium.

Sinus venosus: triangular structure joining right atrium; receives blood from vena cava.

Ventricle opens into conus arteriosus.

Blood circulation:

Arteries carry blood from heart → body.

Veins carry blood from body → heart.

Special venous connections:

Hepatic portal system (liver & intestine).

Renal portal system (kidney & lower body parts).

Blood:

Plasma + blood cells:

RBC (nucleated, haemoglobin)

WBC

Platelets

Function: Transport nutrients, gases, water.

Lymphatic system: Lymph, lymph vessels, lymph nodes.

4. Excretory System

Removes **nitrogenous wastes** (urea); frog is ureotelic.

Components: Pair of kidneys, ureters, urinary bladder, cloaca.

Kidneys: dark red, bean-shaped, near vertebral column.

Each kidney contains uriniferous tubules/nephrons.

Ureters:

Males: ureters act as urinogenital ducts opening into cloaca.

Females: ureters and oviduct open separately into cloaca.

Urinary bladder: thin-walled, ventral to rectum.

6. Nervous and Endocrine System

Nervous system:

Central nervous system: brain + spinal cord.

Peripheral nervous system: cranial & spinal nerves.

Autonomic nervous system: sympathetic & parasympathetic.

Brain: Enclosed in cranium, divided into:

Forebrain: olfactory lobes, cerebral hemispheres, diencephalon.

Midbrain: optic lobes.

Hindbrain: cerebellum, medulla oblongata.

Spinal cord: continuous with medulla oblongata inside vertebral column.

Endocrine glands: Pituitary, thyroid, parathyroid, thymus, pineal body, pancreatic islets, adrenals, gonads.

6. Sense Organs

Organs of touch (sensory papillae), taste (taste buds), smell (nasal epithelium), vision (eyes), hearing (tympanum & internal ears).

Eyes: simple, spherical, in skull orbit.

Tympanum: external ear structure for hearing & balance.

No external ear.

7. Reproductive System

Male:

Pair of testes (yellowish, ovoid) near kidneys.

Vasa efferentia connect testes → kidneys → Bidder's canal → urinogenital duct → cloaca.

Cloaca: common chamber for faeces, urine, sperm.

Female:

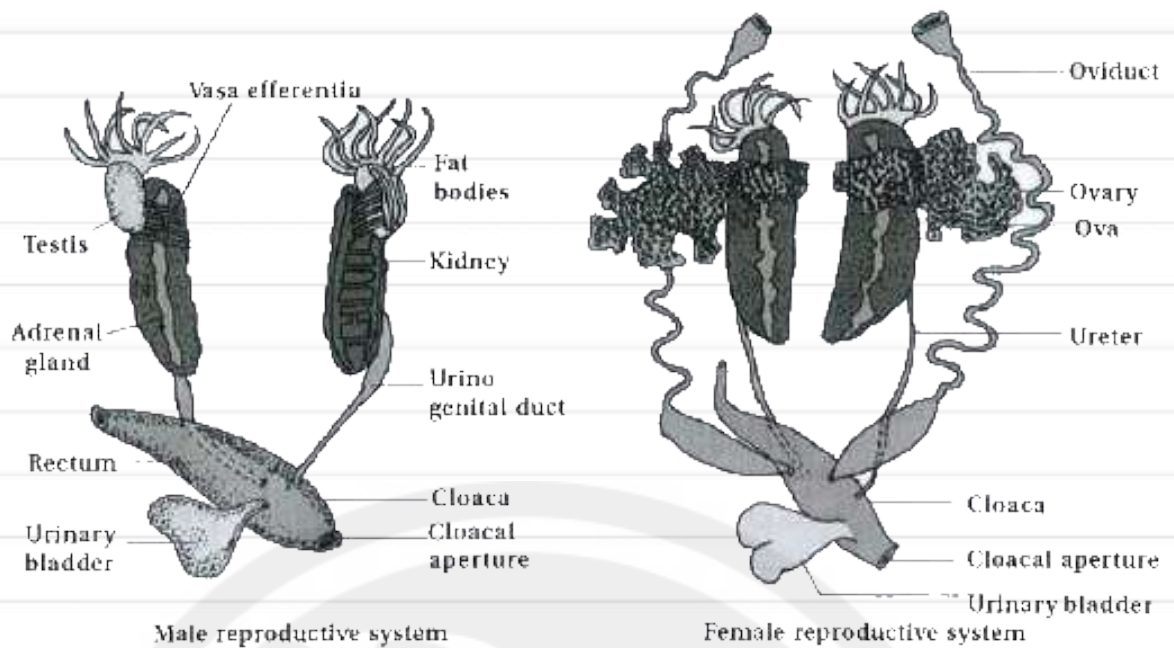
Pair of ovaries near kidneys, no direct connection to kidneys.

Oviducts open separately into cloaca.

Lays 2500–3000 ova at a time.

Fertilisation: external in water.

Development includes larval stage (tadpole) → metamorphosis into adult.



8. Ecological and Economic Importance of Frogs

Control insect population → protect crops.

Maintain ecological balance → important link in food chain/web.

Food source; muscular legs eaten in some countries.